International Flows of Goods, Services and Factors of Production

1MT495 Current Issues in Applied International Economics

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Where We Are

Lecture 1: the accounting

How transactions are recorded: the balance of payments, $CA + KA + NEO = FA$, and the investment position

Today: the map

Why each item of the balance of payments exists – which theory explains which flow, and what the data show

After today you should be able to:

1. match each balance of payments item to the theory that explains it
2. explain why people migrate, why migrants send money home, and how both show up in the accounts
3. read long-run balances across countries and explain typical "current account traps"
4. explain why capital does not flow from rich to poor countries as theory predicts, and why different kinds of capital behave differently

Outline

1. Goods and Services: Why Do Countries Trade?
2. Labour: International Migration
3. Primary Income: The Return on Factors
4. Secondary Income: Remittances and Foreign Aid
5. The Current Account: Sub-Balances and Traps
6. The Capital Account
7. The Financial Account: International Capital Movements
8. Conclusions

Why Do Countries Trade? Two Answers

1. Because they are different

- **Absolute advantage** (Smith): produce more with the same inputs
- **Comparative advantage** (Ricardo): export what you produce at a relatively lower opportunity cost – trade pays even if one country is better at everything
- **Heckscher–Ohlin**: export goods that use your abundant factor intensively
- Explains **inter-industry** trade: cars for coffee

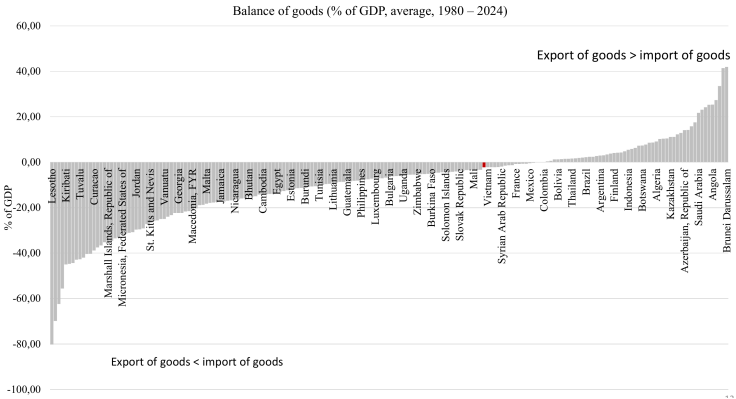
2. Because of economies of scale

- With increasing returns and imperfect competition, **similar** countries trade with each other (Krugman)
- Explains **intra-industry** trade: cars for cars
- Clustering of production in particular places: new economic geography (Krugman 1991)

Leontief paradox (1953): the capital-abundant US exported labour-intensive goods. Accounting for skills, human capital and technology helps explain the result, though no single adjustment resolves every empirical test of Heckscher–Ohlin.

Takeaway: The two answers are complements, not substitutes. Lecture 3 covers the first, lecture 4 the second.

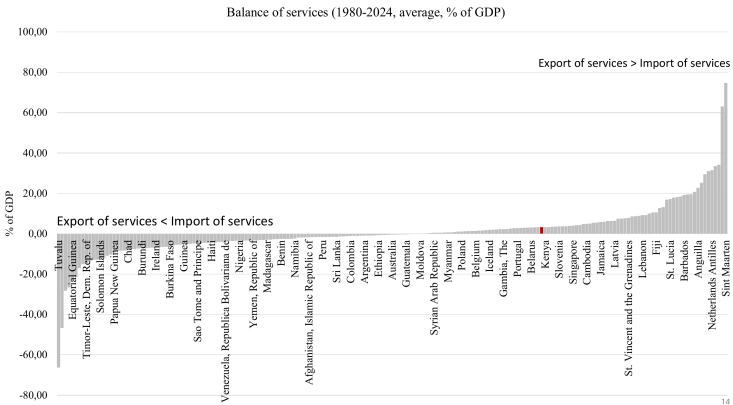
Balance of Goods Around the World, 1980–2024



Takeaway: Large average deficits belong to the poorest economies, war-affected states and small islands; large surpluses to oil and mineral exporters (Saudi Arabia, Angola, Brunei) and manufacturing exporters.

Source: IMF, World Economic Outlook database; averages over available years.

Balance of Services Around the World, 1980–2024



Takeaway: The largest services surpluses belong to small tourist and financial centres (Caribbean islands, Luxembourg, Cyprus); the deepest deficits to the poorest and conflict-affected economies, which import transport and other services.

Source: IMF, World Economic Outlook database; averages over available years.

Reading the Charts: Who Runs Which Balance?

	Large deficit (bottom quartile)	**Large surplus (top quartile)**
Goods	Least developed countries (Lesotho, Nepal, Afghanistan); war-affected states (Liberia, Lebanon); small islands (Cape Verde, Cyprus, Barbados)	Oil and mineral exporters (Saudi Arabia, Angola, Chile, Botswana); manufacturing and re-export hubs (Germany, the Netherlands, Hong Kong, Japan, Korea)
Services	Least developed countries (Lesotho, Chad); some oil exporters (Angola); conflict-affected economies (DR Congo)	Tourist destinations (Maldives, Dominican Republic, Greece); financial centres (Luxembourg, Caribbean islands)

- The middle quartiles mix advanced and emerging economies: there is no simple pattern
- **The Czech Republic** is a manufacturing exporter: its goods and services surplus was about 5% of GDP in the four quarters to mid-2026

Source: O. Šíma, lecture notes; ČNB, balance of payments commentary, 2nd quarter 2026.

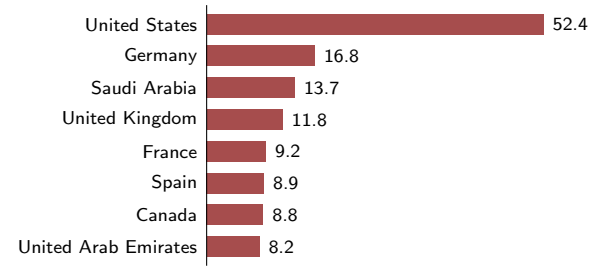
Factors of Production Move Too

Factor	How it moves	Where it appears in the BoP
Labour	Temporary workers (under one year)	Primary income: *compensation of employees*
	Migrants who settle (resident after a year or more)	Secondary income: *personal transfers* (remittances)
Capital	Direct, portfolio and other investment	Financial account (the flow); primary income: *investment income* (the return)
Land, natural resources	Cross-border resource leases; property held by foreign owners	Rent on resources: primary income; non-resident land ownership usually: direct investment*

*Under BPM6, land bought by a non-resident is generally held through a notional resident entity. Diplomatic enclaves are a special case.

Takeaway: Separate the **factor income** from the **financial transaction**: migration itself is not normally recorded as a BoP transaction.

Migration in Numbers, 2024



International migrants by country of destination, million, mid-2024

- **304 million** international migrants in 2024 – **3.7%** of the world population (154 million and 2.9% in 1990)
- 48% are women and girls
- Largest corridor: Latin America and the Caribbean to Northern America (27 million)
- 51.7 million – one in six migrants – were forcibly displaced across borders

Takeaway: Migrants are a small share of the world's population but a large share in some destinations; most migration is not to the poorest-to-richest extremes but within regions and corridors.

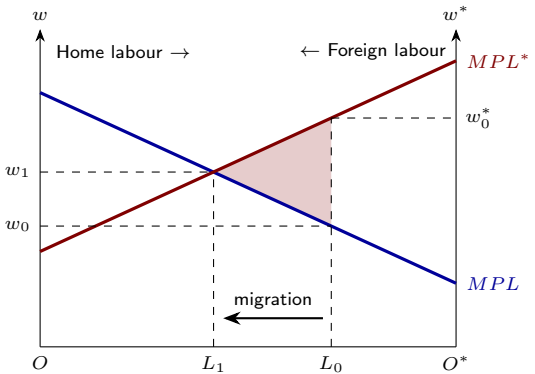
Source: UN DESA, International Migrant Stock 2024: Key Facts and Figures (2025).

Three Families of Migration Theory

Neoclassical	Historical–structural	Networks
Individuals compare wages, distances and the returns to moving. Migration equalises wages.	The economic, social and political environment pulls migrants: segmented labour markets in rich countries demand them.	Migrants stay connected to home. Families diversify risk; networks lower the cost of moving for those who follow.

Takeaway: The theories answer different questions – who moves, why rich countries keep demanding migrants, and why migration persists even when wage gaps shrink.

Neoclassical View 1: The Wage Differential



- World labour OO^* is split between Home (from the left) and Foreign (from the right); wages equal the marginal product of labour
- At L_0 Home has many workers and a low wage w_0; Foreign pays w_0^*
- Workers move until wages are equal at L_1
- **World output rises** by the shaded triangle
- **Distribution:** wages rise at Home and fall in Foreign; Foreign capital and land owners gain

Takeaway: Labour mobility, like trade, raises world output but creates winners and losers in both countries.

Neoclassical Views 2 and 3: Gravity and Human Capital

Gravity model of migration

$$M_{ij} = \alpha \frac{P_i P_j}{D_{ij}}$$

M_{ij} migration from i to j; P population; D_{ij} distance. Migration rises with the size of both countries and falls with distance – the same form as the gravity model of trade (lecture 4).

Migration as an investment in human capital

$$NPV_{ij} = \sum_{t=1}^{T} \frac{(Y_{jt} - Y_{it}) - (C_{jt} - C_{it})}{(1+r)^t} - M$$

Y income and C living costs in the destination j and the origin i; M one-off cost of moving; r discount rate; T years until retirement. Migrate if $NPV_{ij} > 0$.

Worked example

Income gap €10,000 a year, extra living costs €4,000, $r = 5\%$, $T = 30$, $M =$ €20,000:
$NPV = 6{,}000 \times 15.37 - 20{,}000 \approx$ €72,000. For a 55-year-old ($T = 10$): $6{,}000 \times 7.72 - 20{,}000 \approx$ €26,000.

Takeaway: The NPV view explains why migrants are mostly young: they have more years to recoup the cost of moving.

Historical–Structural Theories: The Dual Labour Market

- Migration is shaped by the economic, social, political and cultural environment, not only by individual choice
- Spread in the 1970s–80s with the (non-)integration of post-colonial economies; some versions see rich economies as exploiting cheap labour and resources
- **Dual (segmented) labour market** (Piore 1979): rich economies demand both highly educated and less educated migrants
- Demand persists where jobs have difficult hours, limited progression or unstable conditions; local labour supply may not respond strongly to wage differences

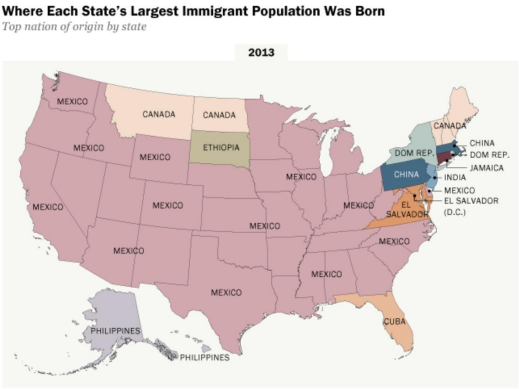


Takeaway: The map shows both segments: farm workers, housekeepers and janitors in most states, college teachers and software developers in others.

Source: Map: Voice of America (2015), "Most common jobs held by immigrants in each US state".

Networks and the New Economics of Labour Migration

- Migrants stay connected to their home country and to those who stayed
- **New economics of labour migration** (Stark and Bloom 1985): migration is a **family** decision to diversify income risk – one member abroad insures the household
- The migrant sends remittances home (secondary income, next section)
- It explains migration **even with a zero wage differential**
- Networks lower the cost of moving for the next migrants: migration follows earlier migration (chain migration)



Takeaway: Mexico is the largest origin in most US states – proximity (gravity) and networks built over decades.

Source: Map: Pew Research Center (2015), "A shift from Germany to Mexico for America's immigrants"; data for 2013.

What Is in Primary Income?

- **Compensation of employees:** wages of border, seasonal and other short-term workers (resident in their home country)
- **Investment income:** the return on the stock of foreign assets and liabilities
  - direct investment: dividends and *reinvested earnings*
  - portfolio investment: dividends and interest
  - other investment: interest on loans and deposits
  - reserve assets: interest earned by the central bank
- **Other primary income:** rent on natural resources; taxes and subsidies on production

The Czech Republic

Net investment income was about -4.6% of GDP in 2025, mostly the profits of foreign-owned firms (net FDI income outflow about 5.2% of GDP) – the price of decades of inward direct investment.

Source: IMF, BPM6, chapter 11; ČNB, Zpráva o vývoji platební bilance 2025.

The Dividend Outflow Problem

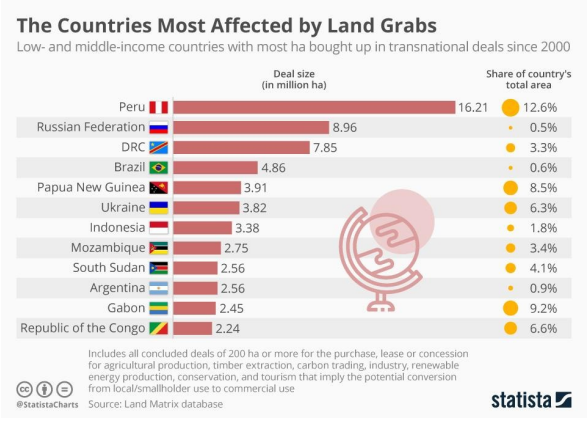
- The large stock of FDI from the 1990s and 2000s generates large income for non-residents: net FDI income outflow of about 5.2% of GDP in 2025
- Two forms:
  1. **Paid dividends:** income payable to foreign investors and usually a cash transfer
  2. **Reinvested earnings:** no dividend cash transfer, but imputed income payable to foreign owners (CA debit) and an offsetting increase in FDI liabilities (FA)
- Thus a primary-income deficit is **not the same as a cash outflow**: reinvested earnings remain available to the domestic enterprise
- **Risk-sharing:** FDI income falls when Czech firms do badly – in a recession the outflow shrinks automatically, unlike interest on debt

Takeaway: The dividend outflow is the price of equity financing: costly in good times, a shock absorber in bad times.

Source: ČNB, Zpráva o vývoji platební bilance 2025.

Land and Natural Resources: "Land Grabbing"

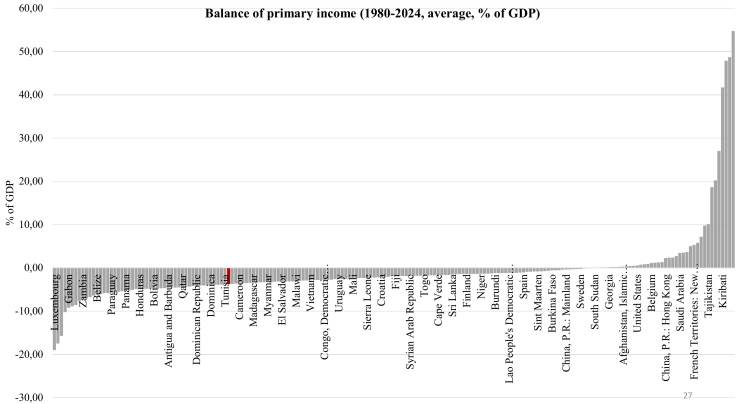
- Land and natural resources in emerging economies are leased or sold to investors from richer countries
- Resource-lease payments are generally **rent** (primary income). Foreign acquisition of land ordinarily creates **direct investment** through a notional resident unit, not a capital-account land sale
- The deals are not always mutually beneficial: *land grabbing* – taking land and resources for rent seeking, often displacing local users



Takeaway: Large-scale land deals raise questions about tenure and local welfare. BoP treatment depends on the *economic ownership arrangement*, not merely the label "land purchase".

Source: Statista, based on the Land Matrix database (large-scale deals since 2000); IMF, BPM6, paragraphs 4.34–4.40.

Balance of Primary Income Around the World, 1980–2024



Takeaway: Primary income balances reflect both returns on foreign asset and liability positions and cross-border labour compensation; a deficit is not automatically a sign of weakness.

Source: IMF, World Economic Outlook database; averages over available years.

Reading the Chart: Primary Income

Deficits

- Emerging recipients of foreign capital (Kazakhstan, Paraguay, Malaysia)
- Financial centres that host foreign equity and debt (Ireland, Luxembourg)
- Advanced economies with foreign-financed mortgage lending (New Zealand)

Surpluses

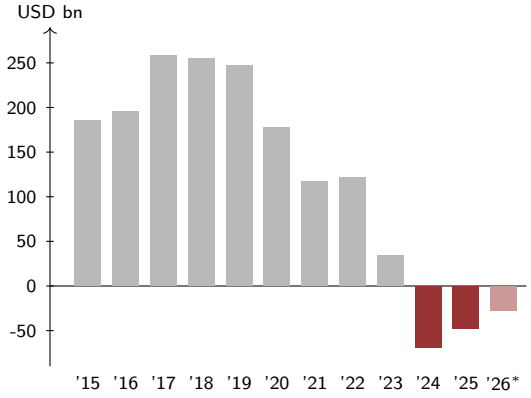
- Creditor economies (Germany, Japan) and financial havens (Bermuda)
- Economies with seasonal or temporary work abroad (Lesotho, Timor-Leste, Tajikistan, Moldova)
- Commodity exporters, surprisingly, show small surpluses: they hold safe, liquid, low-yield assets

The United States

The US ran primary income surpluses for decades despite being a net debtor – it held high-return FDI and owed low-yield debt. That ended in Q4 2023 (next slide).

Source: O. Šíma, lecture notes; U.S. Bureau of Economic Analysis, series IEABCPI via FRED.

The Income Privilege Has Faded



US balance on primary income, USD billion (*first half of 2026)

- The surplus peaked at about $258 bn in 2017
- It turned negative in Q4 2023 and has stayed negative in all but one quarter since: $-\$69$ bn in 2024, $-\$48$ bn in 2025
- Higher rates raised debt-service payments. The net income balance also reflects the large US net liability position and the composition of gross assets and liabilities

Takeaway: The **net primary-income balance** has turned negative. This alone does not prove that the US lost its *return differential*: gross assets and liabilities also differ in size.

Source: U.S. Bureau of Economic Analysis, international transactions; series IEABCPI via FRED (updated 24 September 2026).

Remittances: How They Are Recorded

- Migrants who live in another country for **12 months or more** become **residents** there; money they send home is a **personal transfer** in **secondary income**
- Workers abroad for **less than 12 months** remain residents of their home country; their wages are **compensation of employees** in **primary income**

Component	**Where in the BoP**
Personal transfers	Secondary income
Compensation of employees, net of short-term workers' expenses	Primary income (less goods and services, taxes)
Capital transfers between households	Capital account
Personal remittances = the sum of the three	

Source: IMF, BPM6 (2009), Appendix 5, Table A5.1 (simplified).

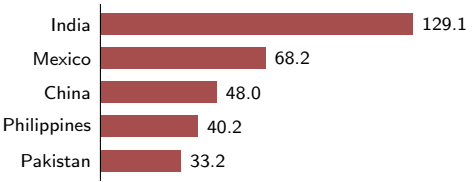
Why Do Migrants Send Money Home?

1. **Pure altruism:** to raise the family's consumption at home
2. **Pure self-interest:** to invest at home – land, a house, a business – or to secure an inheritance
3. **Implicit contract** within the family:
   - insurance – support when the family's income falls
   - repayment – for the cost of the migrant's education or of the move itself
   - exchange – paying relatives to look after children, parents or land
4. **Strategic motive:** raising the income of those at home so that they do not migrate too

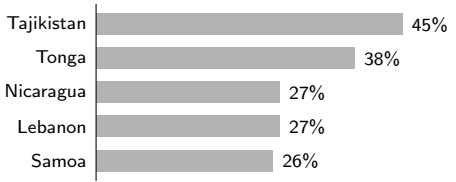
Takeaway: The motive matters for the macro effect: insurance remittances rise when the home economy is hit; investment remittances move with returns at home.

Source: Lucas, R. E. B., & Stark, O. (1985), "Motivations to remit", Journal of Political Economy, 93(5).

Remittances in Numbers



Largest recipients, USD billion, 2024



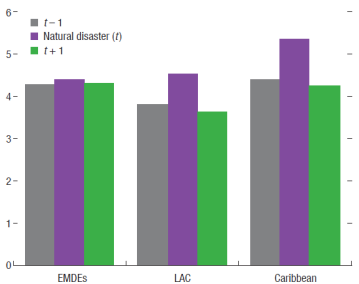
Largest recipients relative to GDP, %, 2024

- Remittances to low- and middle-income countries: about **$685 billion** in 2024 – more than three times official development assistance ($212 billion)
- IFAD's estimate for 2025: $729 billion, with India alone receiving $151 billion

Source: World Bank, Migration and Development Brief 41 (December 2024); IFAD, Sending Money Home (2026); OECD, ODA statistics.

Remittances as Insurance

Figure 5.9. Remittances and Natural Disasters
(Remittances as a percent of GDP)



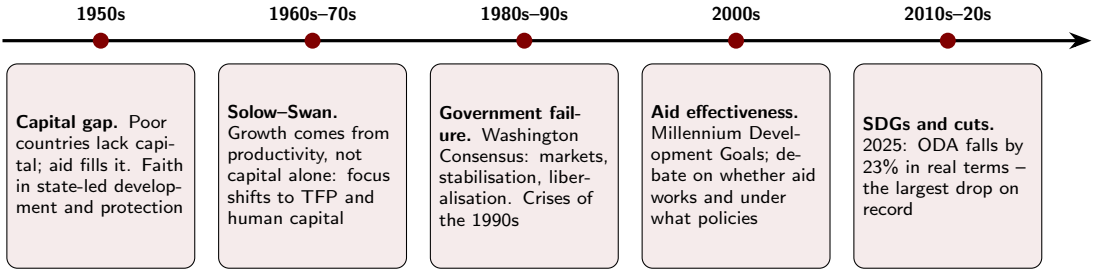
Sources: Emergency Events Database; and IMF staff calculations.
Note: EMDEs = emerging market and developing economies; LAC = Latin America and the Caribbean.

- Remittances as % of GDP the year before (grey), the year of (purple) and the year after (green) a natural disaster
- In the Caribbean they jump from about 4.4% to 5.4% of GDP in the disaster year; in Latin America from about 3.8% to 4.5%
- They fall back the following year
- Unlike private capital, remittances tend to rise when the home economy is hit: they stabilise income

Takeaway: The insurance motive is visible in the data: migrants send more when their families need it.

Source: IMF staff calculations; Emergency Events Database (Figure 5.9 in the original source material; publication to confirm).

Foreign Aid: How Thinking Has Changed



Takeaway: Aid moved from financing capital to building institutions and human capital – and its effectiveness remains contested.

Source: O. Šíma, lecture notes; OECD, preliminary ODA data for 2025 (April 2026).

External Finance for Developing Economies

Figure I.6. Developing economies: sources of external finance, 2012–2022 (Billions of dollars)

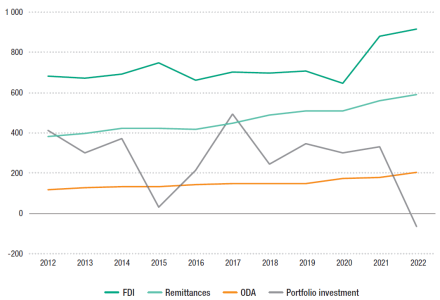
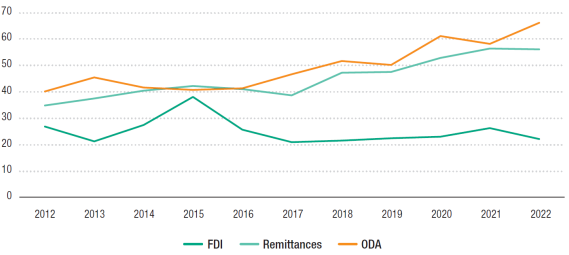


Figure I.11. LDCs: FDI inflows, ODA and remittances, 2012–2022 (Billions of dollars)



Takeaway: For developing economies as a whole, FDI is the largest source and portfolio flows the most volatile (negative in 2022). For the least developed countries, aid (ODA) and remittances each exceed FDI.

Source: UNCTAD, World Investment Report (2023), charts attributed in the original notes (figure numbers to confirm); USD billion, 2012–2022.

Current Issue: The Collapse of Aid in 2025

Official development assistance, 2025

- **$174.3 billion**, down **23.1%** in real terms – the largest annual fall on record
- US aid fell by 56.9% – about three-quarters of the total decline
- Germany became the largest donor for the first time

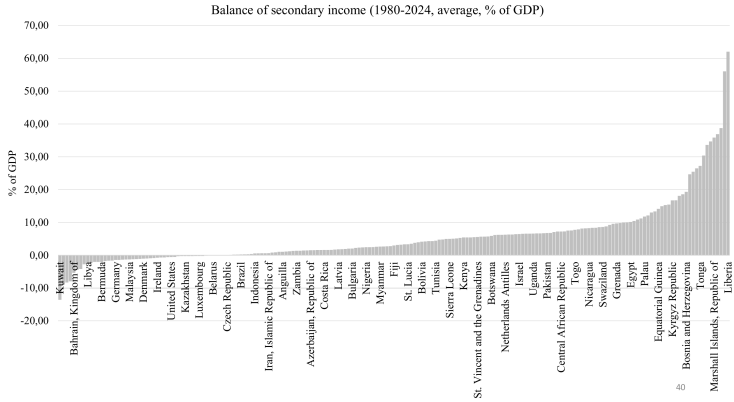
Other flows to developing economies, 2025

- FDI into developing economies: $901 billion ($+2\%$)
- FDI to least developed countries: $+21\%$, to about $43 billion
- Remittances: about $729 billion (IFAD)

Discussion: the poorest countries depend most on aid (previous slide). Can remittances or FDI replace it? Which secondary income and capital account entries change?

Source: OECD, preliminary ODA data (9 April 2026); UNCTAD, World Investment Report 2026; IFAD (2026).

Balance of Secondary Income Around the World, 1980–2024



Takeaway: Deficits: aid donors (Germany), financial hubs and Gulf states with many foreign workers. Surpluses: the poorest and conflict-affected economies (Afghanistan, Yemen, Liberia) that receive aid and remittances.

Source: IMF, World Economic Outlook database; averages over available years.

Two Ways to Read the Current Account

1. Sum of sub-balances

$$CA = \text{goods} + \text{services} + BPI + BSI$$

Each sub-balance has its own drivers – trade, factor income, transfers (this lecture)

2. Saving minus investment

$$CA = S - I = \sum_{\text{sectors}} (S_k - I_k)$$

Gross national saving minus gross capital formation, by sector: firms, financial corporations, government, households (lecture 1)

Takeaway: Both are identities and both are true; they answer different questions – *what* is traded versus *who* saves and invests. The next two slides show both for two countries.

Why Run a Deficit at All? The Intertemporal Approach

The current account as the optimal, forward-looking result of consumption smoothing and investment (Obstfeld and Rogoff 1996).

Two-period small open economy: endowments Y_1, Y_2, world rate r, utility $u(C_1) + \beta u(C_2)$:

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}, \qquad u'(C_1) = \beta(1+r)\,u'(C_2)$$

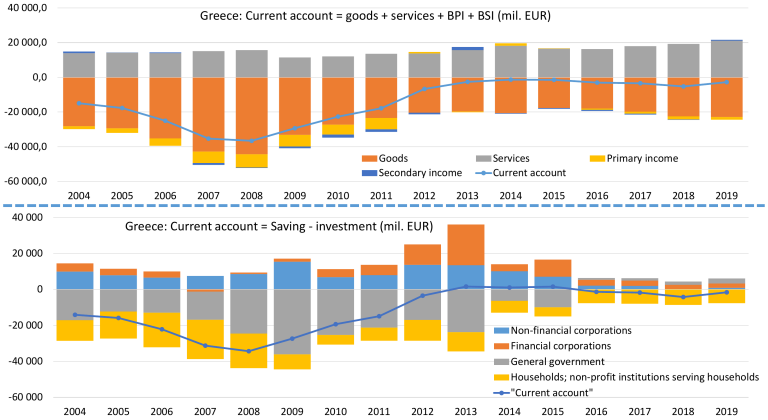
If $\beta(1+r) = 1$, consumption is flat:

$$C_1 = C_2 = \frac{(1+r)Y_1 + Y_2}{2+r}, \qquad CA_1 = Y_1 - C_1$$

- **Temporary** fall in income: borrow – a deficit
- **Permanent** fall: cut consumption – the CA barely moves
- **Investment opportunity:** borrow to invest – a deficit that pays for itself
- **1970s oil shocks:** Saudi Arabia saved its windfalls (surpluses of about $23 bn in 1974 and $41 bn in 1980); oil importers borrowed
- **Limits:** fast-growing China lends rather than borrows – the "allocation puzzle" (Gourinchas and Jeanne 2013); persistent imbalances need other explanations (next slides)

Takeaway: A deficit can smooth a temporary shock or finance investment. Solvency ultimately requires the present value of future trade balances to cover net liabilities, not necessarily a surplus next year. Lecture 5 develops the model.

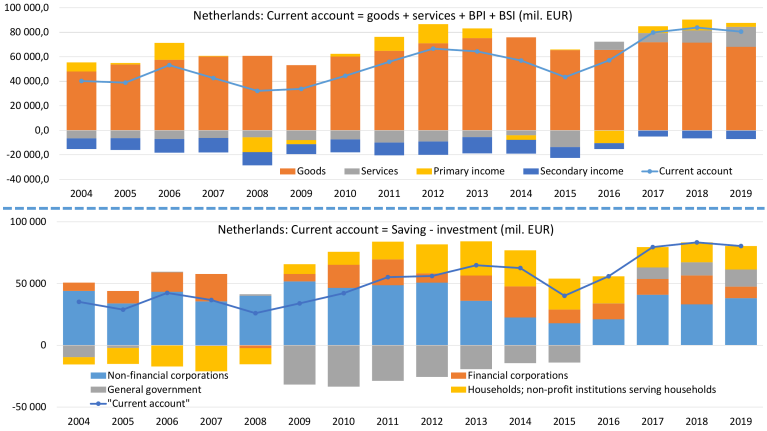
Greece: A Deficit Country, 2004–2019



Takeaway: Top: the deficit of about €35 billion in 2007–08 was a goods deficit, partly offset by services (tourism, shipping). Bottom: the government and households spent more than they saved. The deficit closed by 2013 through the crisis.

Source: Eurostat (EUR million).

The Netherlands: A Surplus Country, 2004–2019



Takeaway: Top: a large goods surplus (€50–75 billion) drives the current account. Bottom: non-financial corporations are the main savers; households (pension saving) add more after 2011; the government was in deficit in 2009–14.

Source: Eurostat (EUR million).

Why Do Some Economies Run Persistent Imbalances? Recurring Patterns

Pattern associated with ...	Current account	Goods	Services	Primary income	Secondary income
... small islands	?	−	+	?	+
... poverty	−	−	− / +	−	+
... demand-led deficit economies	−	−	?	−	?
... creditor/exporter economy	+	+	+	+	−
reserve-currency issuer	−	−	+	+	−
... natural resources	+	+	?	?	−

Illustrative tendencies, not rules. + typical surplus, − typical deficit, ? no a priori sign. Country definitions and periods matter; a single shock can reverse these signs.

Takeaway: Structural characteristics help organise the data, but these are **stylised patterns, not accounting laws or predetermined outcomes**. The saving–investment identity still holds.

Source: O. Šíma, lecture notes.

The Traps Explained (1/2)

- **Geography:** small islands often import goods but earn on tourism and financial services, and may receive remittances (secondary income). Their overall current account can be positive or negative
- **Poverty:** least developed countries run goods deficits, services deficits unless tourism is developed, and primary income deficits on earlier capital imports; aid and remittances give a secondary income surplus. Few countries have left least-developed status
- **Demand-led deficit economies:** persistent spending above national income can accompany goods deficits; previous borrowing may add net interest payments. Government deficits, housing booms and private saving all matter

The Traps Explained (2/2)

- **Creditor/exporter economies:** competitive exports and accumulated foreign assets can create goods, services and investment-income surpluses; transfers abroad may reduce the overall balance
- **Reserve-currency issuer:** the most stable economies can run lasting deficits. The currency tends to be strong because the world demands assets in it, so net exports are in deficit; the primary income balance is higher than its net debt would suggest, because it borrows cheaply in its own currency and earns more on its foreign assets (the "exorbitant privilege") – although for the US this surplus turned into a small deficit after 2023
- **Natural resources:** large goods surpluses may be offset by imported services, income owed to foreign investors and transfers by migrant workers. Income earned on sovereign wealth funds can work in the opposite direction

The Capital Account

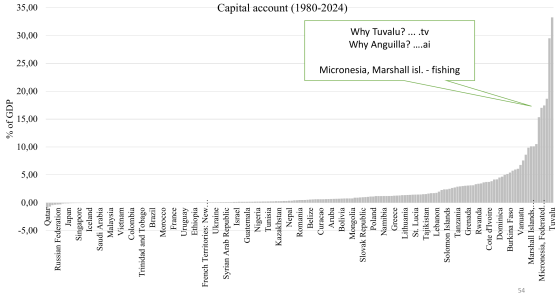
Capital transfers:

- EU structural and investment funds
- Debt forgiveness
- Household capital transfers (inheritances)

Non-produced, non-financial assets:

- Emission allowances
- Purchases of land for foreign diplomatic enclaves (special case)
- Sales of brands, trademarks, domain names

Under BPM6, a migrant's change of residence is not a transaction: the migrant's assets are recorded as other changes in volume.



Discussion

Tuvalu (.tv), Anguilla (.ai), and Pacific fisheries stand out. Where do these receipts go? An outright transfer of ownership of a qualifying *non-produced asset* may be capital account; an ongoing licence, royalty or access fee is classified according to the contractual rights (often services or rent).

Source: IMF, World Economic Outlook database; averages over 1980–2024; IMF, BPM6, chapters 9 and 13.

Theory: Capital Should Flow to Poor Countries

- In the neoclassical (Solow–Swan) model the marginal product of capital falls as capital per worker rises:

$$y = Ak^{\alpha} \quad \Rightarrow \quad MPK = \alpha A k^{\alpha-1} = \alpha A^{1/\alpha}\, y^{(\alpha-1)/\alpha}$$

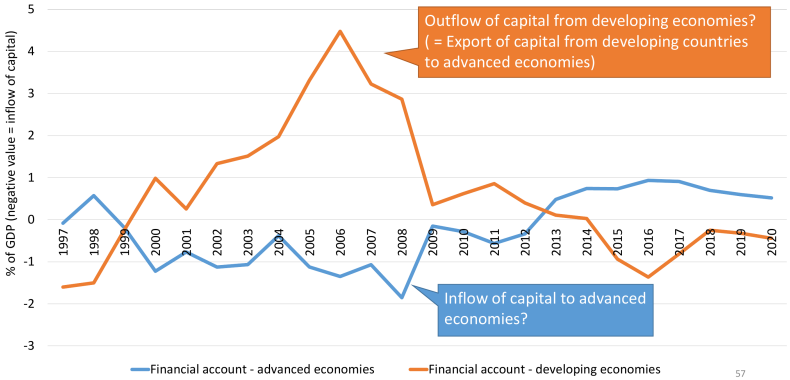
- So capital should flow from rich, capital-abundant economies to poor, capital-scarce ones
- **Lucas (1990):** *if TFP is equal*, with $\alpha = 0.4$ and US output per worker 15 times India's, the MPK in India would be

$$15^{(1-\alpha)/\alpha} = 15^{1.5} \approx 58 \text{ times the US level}$$

- Such predicted returns would induce enormous capital inflows. They are not observed.

Takeaway: The benchmark predicts large returns to capital scarcity. Differences in productivity, institutions, risk and financial development help explain why observed flows are far smaller.

Reality: Capital Flows "Uphill"



Takeaway: From 2000 to 2013 developing economies were net *lenders* (financial account positive, up to about 4.5% of GDP in 2006) and advanced economies net borrowers – capital flowed from poor to rich.

Source: IMF, World Economic Outlook database. Financial account in % of GDP; positive = net lending (net capital outflow).

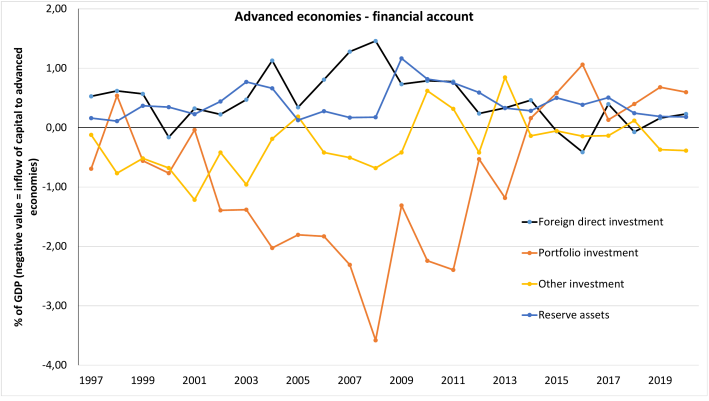
Why Uphill? Capital Is Not One Thing

Type	Main motive	Volatility	For the recipient
Direct investment	Control: markets, low costs, resources	Generally lower	Equity shares risks and profits; intra-group debt is also possible
Portfolio equity	Diversification, returns	High	Shares risk, but can leave quickly
Portfolio debt	Yield, safety, liquidity	High	Rollover and currency risk
Other investment	Bank lending, trade credit, deposits	Highest (short-term)	First to run in a crisis
Reserve assets	Insurance; exchange-rate policy	Policy-driven	Safe, liquid, low yield

- **Explanations of the paradox:** lower risk-adjusted returns – weak institutions, sovereign risk (Alfaro, Kalemli-Ozcan and Volosovych 2008); **official** flows – emerging central banks buying reserves in advanced economies, while private capital still flows to poorer countries

Takeaway: Net flows hide different gross transactions: some emerging markets receive FDI while acquiring safe foreign assets and reserves; the pattern varies by country and period.

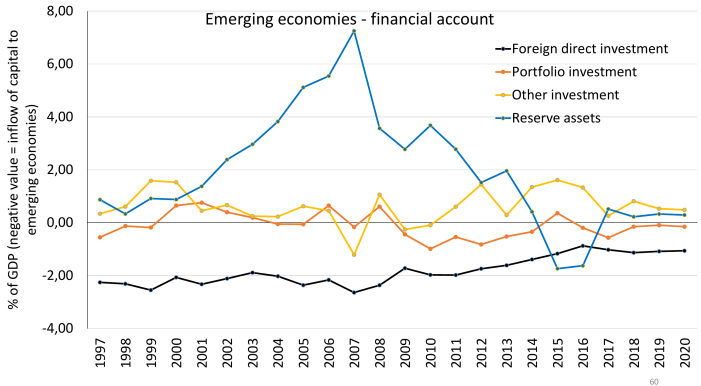
Advanced Economies: Financial Account by Type



Takeaway: Advanced economies are net exporters of FDI (black, positive) and, until 2013, large net importers of portfolio capital (orange, down to about -3.6% of GDP in 2008) – the world's demand for their safe assets.

Source: IMF, World Economic Outlook database; % of GDP; negative = net inflow.

Emerging Economies: Financial Account by Type



Takeaway: Emerging economies receive a steady inflow of FDI (black, about -2% of GDP) and recycle it into reserves (blue), which peaked at about 7% of GDP in 2007. In 2015–16 reserves were sold, above all by China.

Source: IMF, World Economic Outlook database; % of GDP; negative = net inflow.

Three Trends in International Capital Movements

a) **Accumulation of reserve assets**, mainly by emerging economies

b) **Capital flowing from emerging to advanced economies**, above all through portfolio investment in safe assets

c)

1. **Persistent FDI inflows** to many emerging economies, alongside potentially volatile financial flows

Advanced economies

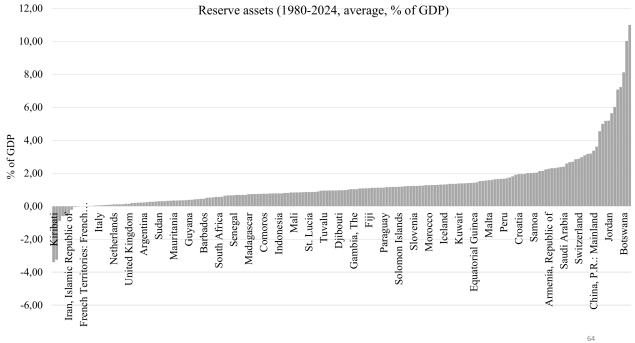
invest their savings abroad in less liquid, riskier, higher-return assets (FDI) and borrow through portfolio debt

Emerging economies

invest abroad in liquid, safe, low-return assets (central bank reserves) because they are exposed to liquidity shocks, and import capital as FDI

a) Why Do Central Banks Accumulate Reserves?

- **Precautionary:** a buffer against currency crises and sudden stops – after Asia 1997 (rule of thumb: cover short-term external debt, Greenspan–Guidotti)
- **Mercantilist:** buying foreign currency keeps the domestic currency weak and exports competitive
- **Transactions:** smoothing foreign payments; government payments
- **Financial stability:** foreign-currency liquidity for banks



Global FX reserves: **USD 13.2 trillion** (Q2 2026).

Takeaway: Botswana, Jordan and China accumulated reserves of several per cent of GDP a year on average; most countries add about 1%.

Source: IMF, World Economic Outlook database, (average annual change, % of GDP); IMF, COFER (Q2 2026).

The Czech Case: Capital Flows and the Koruna Commitment, 2013–2017

- **7 November 2013:** the ČNB commits to prevent the koruna from appreciating beyond CZK 27/EUR
- **From mid-2015:** inflows betting on the exit force heavy intervention – EUR 75 bn in total by March 2017
- Non-residents took 47% of government bond auctions in 2015–16 and 55% in the first half of 2017; short-term yields turned negative from August 2015. By 2025 non-residents held 25% of koruna government bonds (37% in 2019)
- **6 April 2017:** exit with immediate effect; the koruna appreciated gradually, without a gap

Why the exit was orderly

Positions were large but two-sided: exporters and investors sold the koruna as speculators bought it, so the central bank was not the only absorber.

Takeaway: A credible one-way bet attracts large gross inflows into local bonds; reserves absorb them, and the exit tests liquidity and positioning.

Source: ČNB, Inflation Report II/2017, Box 1; Inflation Report III/2017, Box 2; ČNB, Zpráva o vývoji platební bilance 2025 (Ministry of Finance data).

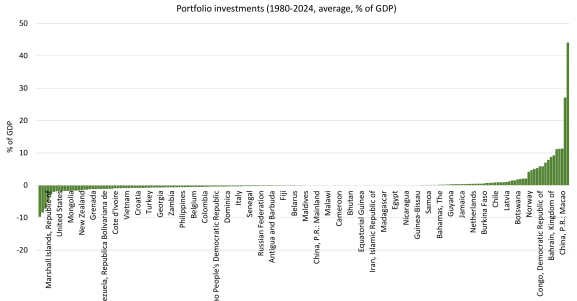
b) The Demand for Safe Assets

What makes an asset safe?

- The issuer's institutional quality and rating
- A market large enough to be liquid
- A record of behaving safely in past crises
- In short: its value holds when shocks hit

Risk on, risk off:

- In calm times private and public assets are close substitutes
- When risk aversion rises, safe assets are mainly government bonds of advanced economies – the US, Germany, Japan, Switzerland

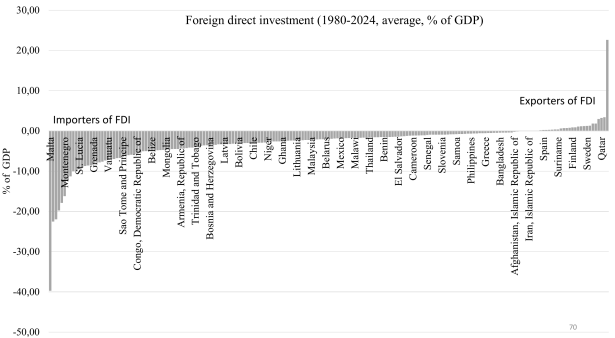


Takeaway: Emerging economies' saving flows into advanced-economy bonds: the US has run one of the largest average portfolio inflows (left), while Norway's oil fund and Asian surplus economies are large buyers abroad (right).

Source: IMF, World Economic Outlook database, (portfolio investment, average, % of GDP; negative = net inflow).

c) Foreign Direct Investment

- **Motives** (Dunning's OLI framework): firms invest abroad when they own an advantage (O), the location offers something (L) – markets, low costs, resources – and it pays to keep it in-house (I)
- **Return to investors:** profits appear in primary income; equity also shares downturn losses
- **Usually relatively stable:** ownership stakes are harder to unwind than short-term debt, though intra-group loans can reverse
- Includes equity *and* intra-group debt; may bring management, technology and market access
- **Caveat:** part of measured FDI passes through holding companies in financial centres (Malta, Luxembourg, the Netherlands)



Global FDI 2025: **$1.6 trillion** ($+6\%$).

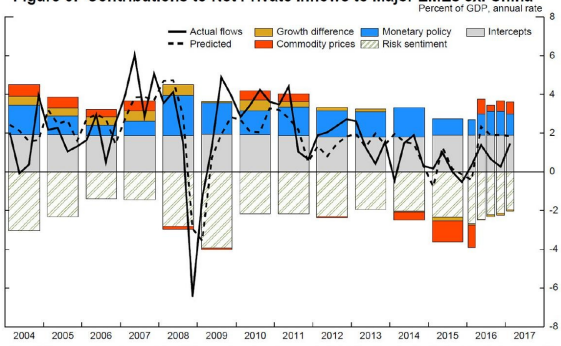
Source: IMF, World Economic Outlook database, (negative = net importer of FDI); UNCTAD, World Investment Report 2026.

What Drives Portfolio Flows? Push and Pull

- **Push (global):** interest rates in advanced economies, global risk aversion, global growth
- **Pull (domestic):** interest differential, growth, fundamentals, institutions
- **Risk on:** debt capital flows into emerging markets; **risk off:** it flows back to advanced economies
- **Herding** amplifies both
- Also: financial openness, exchange-rate regime, capital flow measures

Compare with FDI, which responds to long-run location factors.

Figure 5: Contributions to Net Private Inflows to Major EMEs ex. China



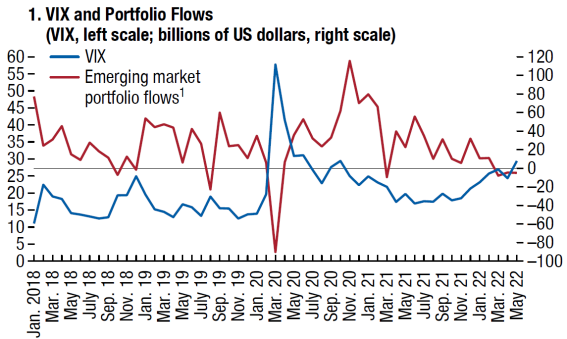
Note: EMEs are emerging market economies; GDP is gross domestic product. Countries included are Argentina, Brazil, Chile, Colombia, Czech Republic, Hungary, India, Indonesia, Israel, Malaysia, Mexico, Philippines, Poland, Romania, Russia, South Africa, South Korea, Taiwan, Thailand, and Turkey.
Source: International Monetary Fund; Haver Analytics; Federal Reserve Board staff estimates.

Takeaway: Model decomposition for major emerging economies (excluding China): risk sentiment (hatched) and monetary policy (blue) explain most of the swings, including the collapse of inflows in 2008.

Source: Federal Reserve Board staff estimates; data: IMF and Haver Analytics (publication of chart to confirm); Calvo, Leiderman and Reinhart (1993) for push and pull.

Risk Off: The VIX and Emerging-Market Flows

Figure 1.12. Capital Flows to Emerging Market and Developing Economies and the VIX

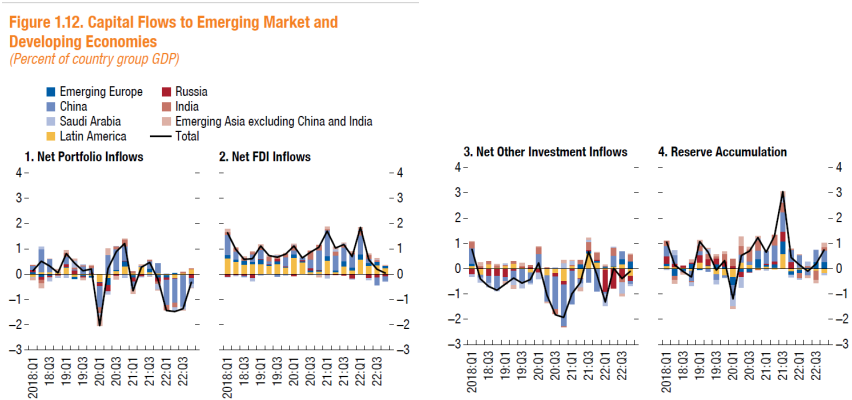


- The **VIX** measures expected volatility of the S&P 500 over the next 30 days, from option prices – a gauge of fear
- March 2020: the VIX jumped to nearly 60 and emerging-market portfolio flows turned to an outflow of about $90 billion in a month
- Flows recovered within months as central banks eased

Takeaway: Portfolio flows to emerging markets move against global risk aversion: a "push" factor no emerging economy controls.

Source: IMF chart attributed as Figure 1.12 in original notes (publication to confirm); VIX and portfolio flows.

Capital Flows to Emerging Economies by Type, 2018–2022



Takeaway: Portfolio flows swing hardest (about -2% of GDP in early 2020, outflows from China in 2022); FDI stays positive; reserve accumulation spiked in 2021 with the IMF's SDR allocation.

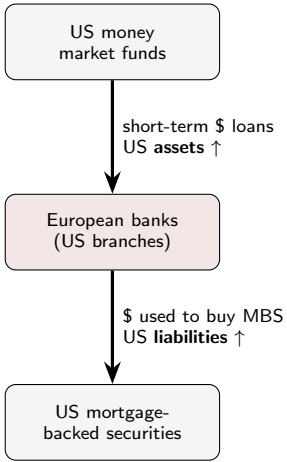
Source: IMF chart attributed as Figure 1.12 in original notes (publication to confirm); % of country-group GDP.

Gross vs. Net Capital Flows: The 2008 Example

Before 2008 the euro area's current account was close to balance, but **gross flows** with the US exploded:

- US money market funds lent dollars to European banks $\Rightarrow$ **US foreign assets** rise (claims on European banks)
- European banks used the dollars to buy US mortgage-backed securities $\Rightarrow$ **US foreign liabilities** rise
- Net flow ≈ 0; gross positions large and maturity-mismatched (short-term funding, long-term assets)

When mortgage values fell in 2007–08, the money funds pulled back: a **sudden stop** in dollar funding for European banks, contained only by the Federal Reserve's dollar swap lines with other central banks.



net flow ≈ 0, gross flows large

Gross Flows: Surges, Stops, Flight and Retrenchment

Gross **inflows** (by foreigners)	**Surge** sharp rise	**Stop** sharp fall
Gross **outflows** (by residents)	**Flight** sharp rise	**Retrenchment** sharp fall

Forbes and Warnock (2012)

- Episodes are driven mainly by **global factors** – risk aversion, global growth – and spread through contagion; domestic factors matter less
- **Sudden stop** (Calvo 1998): an abrupt fall in net inflows forces the current account to adjust – a crisis when debt is short-term and in foreign currency

What a central bank monitors

Gross inflows and outflows by type and investor; short-term external debt relative to reserves; currency mismatches; non-resident holdings of local-currency bonds; rollover of bank funding.

The Policy Toolkit for Capital Flows

Instrument	What it does	Examples and caveats
Exchange rate flexibility, monetary policy	First line of defence: the exchange rate absorbs shocks	Fear of floating where debt is in foreign currency
FX reserves and intervention	Self-insurance against stops; smooths disorderly moves	Asia's reserve build-up after 1997; costly to hold
Macroprudential measures (MPMs)	Limit FX lending, currency mismatches and leverage at home	Limits on FX mortgages in Central and Eastern Europe after 2008
Capital flow management measures (CFMs)	Tax or limit inflows or outflows by residence	Chile's reserve requirement (1991–98); Brazil's tax on portfolio inflows (from 2009)

IMF Institutional View (2012, reviewed 2022)

CFMs should not substitute for needed macroeconomic adjustment. Since 2022, **pre-emptive** CFM/MPMs on debt inflows may be appropriate where stocks of foreign currency debt create systemic risk – targeted, temporary and transparent. The Integrated Policy Framework (Basu et al. 2020) studies how the four tools work together.

Conclusions

- Every balance of payments item has its own theory: trade for goods and services, factor mobility for primary income, migration and aid for secondary income, saving and investment for the current account, portfolio and firm theory for the financial account
- **Migration** is explained by wage gaps, distance and human-capital returns, by segmented labour markets, and by family strategies; it shows up as compensation of employees and remittances
- **Remittances** ($685 billion in 2024) far exceed aid and stabilise income; **aid** fell by 23% in 2025
- Lasting **current account imbalances** often reflect structural "traps": geography, poverty, currency status, natural resources
- **Capital flows uphill** in net terms because capital is heterogeneous: emerging economies import FDI and export safe-asset demand and reserves
- **Net hides gross:** abrupt reversals in cross-border funding can trigger crises even when the current account is near balance; policy tools include reserves, exchange-rate flexibility and prudential measures

Next lecture: neoclassical trade theory – why countries export what they do, and who gains.

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